

St George Girls High School

Trial Higher School Certificate Examination

2010



Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Begin each question on a new booklet
- Write your student number on each page
- All necessary working must be shown.
- Diagrams are not to scale.
- Board-approved calculators may be used.
- The mark allocated for each question is listed at the side of the question.

Total Marks – 120

- Attempt ALL questions.
- All questions are of equal value.

Students are advised that this is a Trial Examination only and does not necessarily reflect the content or format of the Higher School Certificate Examination.

Question 1 – (12 marks)

Marks

- a) Given $g(x) = \begin{cases} 1 - x^2 & x \leq 2 \\ 2^x & x > 2 \end{cases}$ evaluate $g(3) + g(2)$ 2
- b) Factorise fully $9x^3 - 9$ 2
- c) Convert 108° to radians giving your answer in terms of π . 2
- d) Solve $\frac{x-5}{3} - \frac{x+1}{4} = 5$ 2
- e) Solve and graph $|5x - 3| < 7$ 2
- f) Expand and simplify $(\sqrt{2} + \sqrt{6})^2$ 2

Question 2 – (12 marks)

Marks

a) Differentiate

(i) $(1 - 5x)^5$ 2

(ii) $3x e^{3x}$ 2

(iii) $\log_e(\cos x)$ 2

b) Evaluate $\int_0^{\frac{\pi}{6}} \cos 2x \, dx$ 2

c) Find:

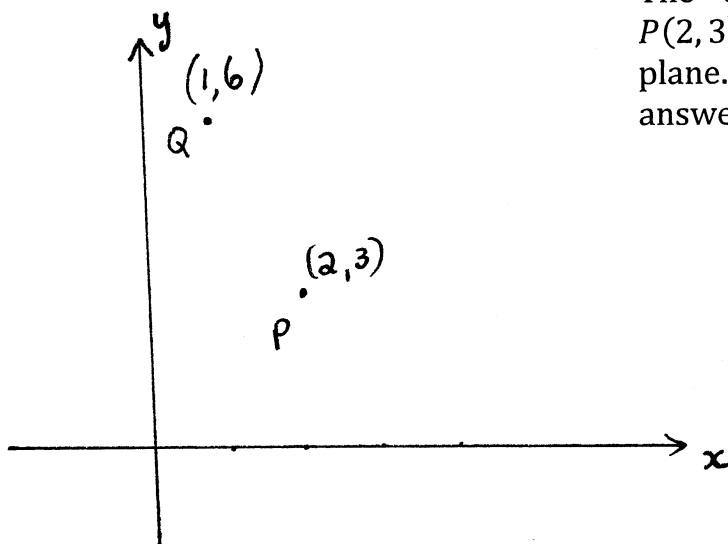
(i) $\int \sqrt{x} \, dx$ 2

(ii) $\int 2x^2 e^{x^3} \, dx$ 2

Question 3 – (12 marks)

Marks

a)



The diagram shows two points $P(2,3)$ and $Q(1,6)$ on the number plane. Copy the diagram into your answer booklet.

- (i) Find the coordinates of M the midpoint of PQ . 1
- (ii) Show that the equation of the perpendicular bisector of PQ is $x - 3y + 12 = 0$ 2
- (iii) Find the coordinates of the point R that lies on the y -axis and is equidistant from P and Q . 1
- (iv) The point S lies on the intersection of the line $y = 6$ and the perpendicular bisector, $x - 3y + 12 = 0$. Find the coordinates of S and mark the position of S on your diagram. 2
- (v) Find the area of the triangle PQS . 2

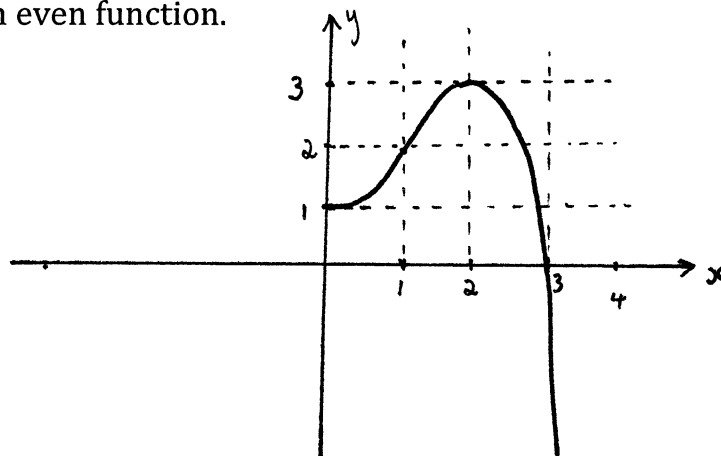
b) Differentiate the following:

- (i) $y = x \sin 3x$ 2
- (ii) $y = \frac{\ln x}{x^2}$ 2

Question 4 – (12 marks)

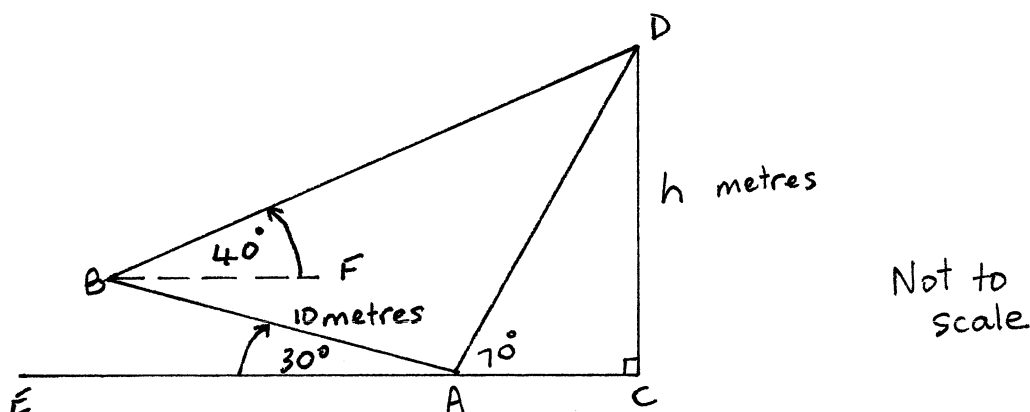
Marks

- a) The following diagram shows the graph of $y = f(x)$ for $x \geq 0$. It is known that $f(x)$ is an even function.



- (i) Copy the diagram into your answer booklet and complete the graph for $x < 0$. 2
- (ii) On a separate diagram sketch a graph of $y = f'(x)$ 2

b)



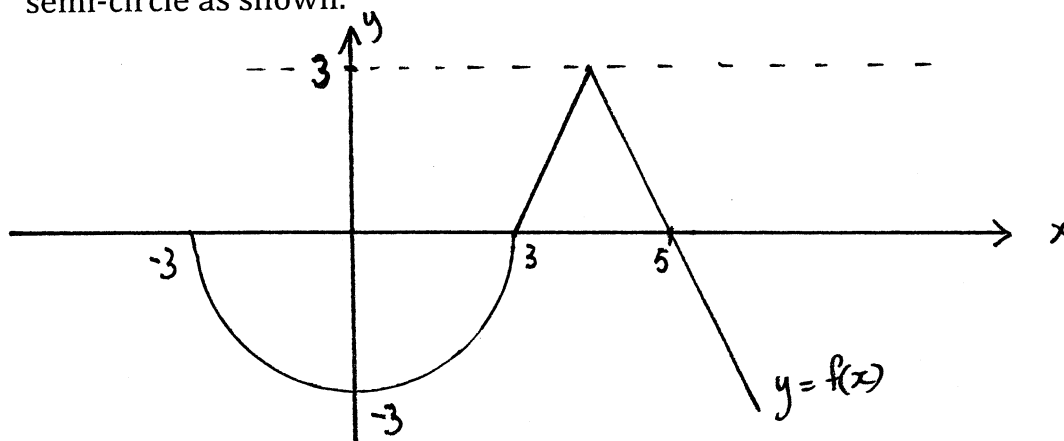
The top of a vertical tower CD is observed at an angle of elevation 70° from A . After walking 10 metres up a ramp AB , inclined at an angle of 30° to the horizontal, the top of the tower is now observed at an angle of elevation of 40° .

- (i) Find, with reasons the size of $\angle ABD$, and of $\angle BDA$. 4
- (ii) Find the length of AD . 2
- (iii) Hence find the height of the tower correct to one decimal place. 2

Question 5 – (12 marks)

Marks

- a) The graph of a function $y = f(x)$ consists of two straight line sections and a semi-circle as shown.



Find the exact value of $\int_0^5 f(x) dx$

2

- b) Let $\log_a 2 = x$ and $\log_a 3 = y$

Find an expression for $\log_a 24$ in terms of x and y .

2

- c) Evaluate

$$\sum_{n=3}^8 (2n^2 - 3)$$

2

- d) Find:

(i) $\int_1^4 x \sqrt{x} dx$

2

(ii) $\int \frac{5x^2}{x^3+1} dx$

2

(iii) $\int_0^2 e^x + e^{-x} dx$

2

Question 6 – (12 marks)

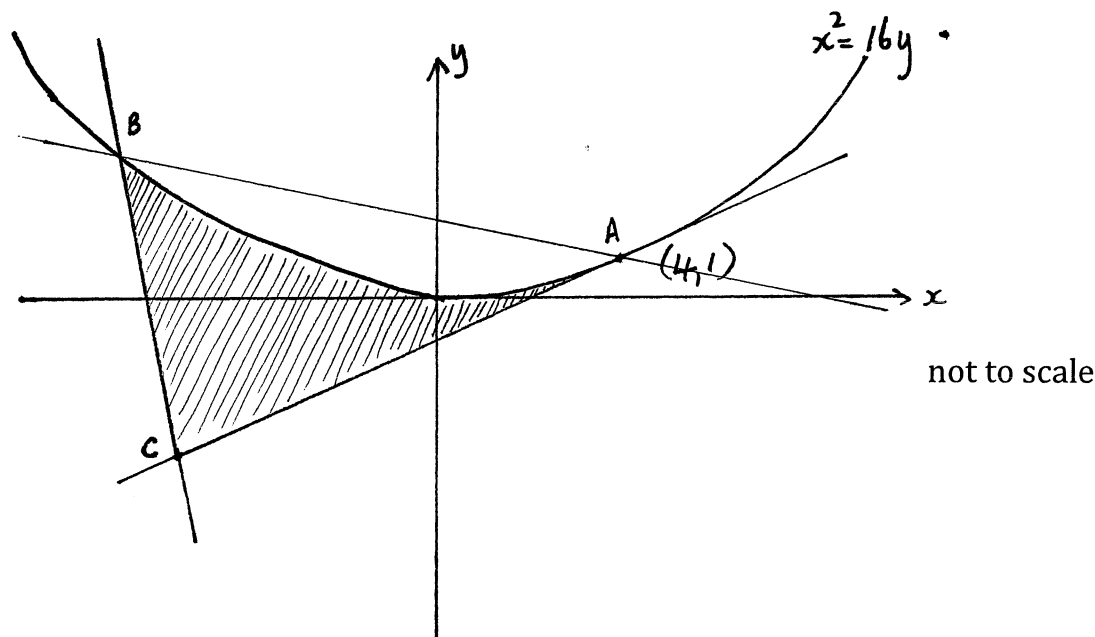
Marks

- a) Solve the following equation for x

3

$$e^{2x} + 5e^x - 14 = 0$$

b)



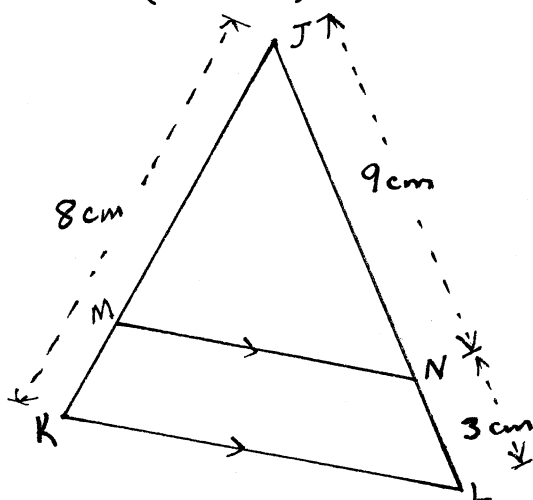
The diagram shows the graph of the parabola $x^2 = 16y$. The points $A(4, 1)$ and $B(-8, 4)$ are on the parabola, and C is the point where the tangent to the parabola at A intersects the directrix.

- (i) Write down the equation of the directrix of the parabola $x^2 = 16y$. 1
- (ii) Find the equation of the tangent to the parabola at A . 2
- (iii) Show that C is the point $(-6, -4)$. 1
- (iv) Given that the equation of AB is $y = 2 - \frac{x}{4}$
 - (α) Find the perpendicular distance of C to the line AB . 1
 - (β) Find the area bounded by the line AB and the parabola $x^2 = 16y$. 2
- (v) Hence or otherwise find the shaded area bounded by the parabola, the tangent at A and the line BC . 2

Question 7 – (12 marks)

Marks

a)



In the diagram triangle JKL is shown $MN \parallel KL$, $JK = 8$ cm, $JN = 9$ cm and $NL = 3$ cm.

(i) Prove that $\triangle JMN$ is similar to $\triangle JKL$.

2

(ii) Find the length of MK .

2

b) Consider the function $y = 2^x$.

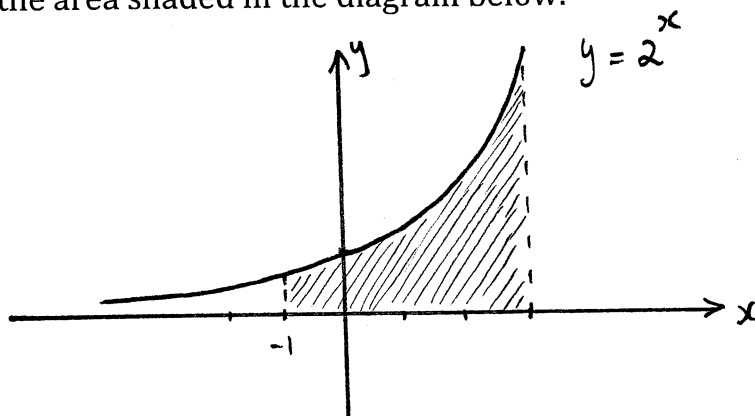
x	-1	0	1	2	3
2^x					

(i) Copy and complete the above table in your writing booklet.

1

(ii) Using Simpson's Rule with these five function values, find an estimate for the area shaded in the diagram below.

2



Question 7 (cont'd)

Marks

- c) A particle is moving in a straight line according to the formula $x = t^2 - 2t + 7$, where t is time in seconds, and x is the displacement from the origin O in metres. Find:
- (i) the initial displacement of the particle. 1
 - (ii) the time when the particle is stationary. 2
 - (iii) the total distance travelled by the particle in the first 4 seconds. 2

Question 8 – (12 marks)

Marks

- a) (i) Write down the discriminant of $3x^2 + 2x + k$. 1
- (ii) For what values of k does $3x^2 + 2x + k = 0$ have real roots? 1
- b) A ball is dropped from a height of 3 metres onto a hard floor and bounces. After each bounce, the maximum height reached by the ball is 80% of the previous maximum height. Thus, after it first hits the floor it reaches a height of 2.4 metres before falling again and after the second bounce, it reaches a height of 1.92 metres before falling again.
- (i) What is the maximum height reached on the third bounce? 1
- (ii) What kind of sequence is formed by the successive maximum heights? 1
- (iii) What is the total distance travelled by the ball from the time it was first dropped until it eventually comes to rest on the floor? 2
- c) (i) Sketch the graph of $y = x^2 - 6$ and label all intercepts with the axes. 2
- (ii) On the same set of axes, carefully sketch the graph of $y = |x|$ 1
- (iii) Find the x coordinates of the two points where the graphs intersect. 2
- (iv) Hence solve the inequality $x^2 - 6 \leq |x|$ 1

Question 9 – (12 marks)

Marks

- a) Penny is saving for an overseas trip. At the beginning of each month she deposits \$400 into a bank account which pays 6% p.a. calculated monthly.

(i) How much will she have after 2 years?

2

(ii) Penny actually needed \$16 000 for her trip. How much should she have deposited each month into this account for her to have reached her goal in 2 years?

2

- b) Given $y = 2 \cos 3x$ write down

(i) the amplitude of this graph.

1

(ii) the period.

1

(iii) sketch the graph for $0 \leq x \leq 2\pi$.

2

- c) (i) Show that the volume of the solid formed when $y = \tan 2x$ is rotated about the x -axis between $x = 0$ and $x = \frac{\pi}{6}$ is given by

2

$$V = \pi \int_0^{\frac{\pi}{6}} (\sec^2 2x - 1) dx$$

(ii) Find the exact volume of the solid.

2

Question 10 – (12 marks)

Marks

a) For the parabola $(x + 3)^2 = 4y - 14$ find the following:

- | | |
|--------------------------------------|---|
| (i) the vertex. | 1 |
| (ii) the focus. | 2 |
| (iii) the equation of the directrix. | 1 |

b) A cam is formed with a cross-section as shown in the figure.

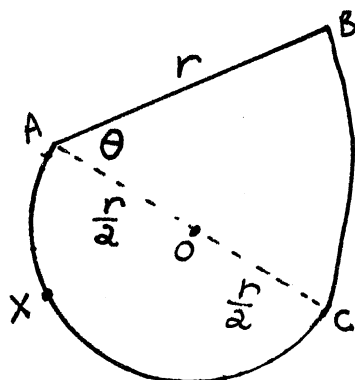


figure not to scale

The cross-section consists of a semicircle AXC , centre O and radius $\frac{r}{2}$, and a sector ABC of radius r and centre A and angle θ .

- | | |
|--|---|
| (i) What is the perimeter (P) of the cam $ABCX$ in terms of r and θ ? | 2 |
| (ii) If the area of the cross section of the cam is 1 square unit, by first finding θ show that the perimeter P is given by | 3 |

$$P = \frac{2}{r} + r \left(1 + \frac{\pi}{4} \right)$$

- | | |
|--|---|
| (iii) Show that the least perimeter occurs when $r^2 = \frac{8}{\pi+4}$ and calculate the value of θ when the perimeter is least. | 3 |
|--|---|

End of Paper

Solutions - Mathematics Trial 2010

Question 1.

(a) $g(3) + g(2) = 2^3 + 1 - 2^2$
 $= 5$

(b) $9x^3 - 9 = 9(x^3 - 1)$
 $= 9(x-1)(x^2 + x + 1)$

(c) $108^\circ = \frac{3\pi}{5}$ radians

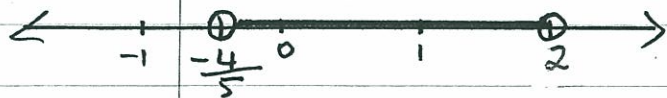
(d) $\frac{x-5}{3} \times 12 - \frac{x+1}{4} \times 12 = 5 \times 12$
 $4(x-5) - 3(x+1) = 60$
 $4x - 20 - 3x - 3 = 60$
 $x - 23 = 60$
 $x = 83$

(e) $|5x-3| < 7$

$$-7 < 5x-3 < 7$$

$$-4 < 5x < 10$$

$$-\frac{4}{5} < x < 2$$



(f) $(\sqrt{2} + \sqrt{6})^2 = 2 + 2\sqrt{12} + 6$
 $= 8 + 2 \times 2\sqrt{3}$
 $= 8 + 4\sqrt{3}$

Question 2.

(a) (i) $y = (1-5x)^5$
 $y' = 5(1-5x)^4 \times -5$
 $= -25(1-5x)^4$

(ii) $y = 3x e^{3x}$
 $y' = 3e^{3x} + 3x \times 3e^{3x}$
 $= 3e^{3x} + 9xe^{3x}$
 $= e^{3x}(3+9x)$
 $= 3e^{3x}(1+3x)$

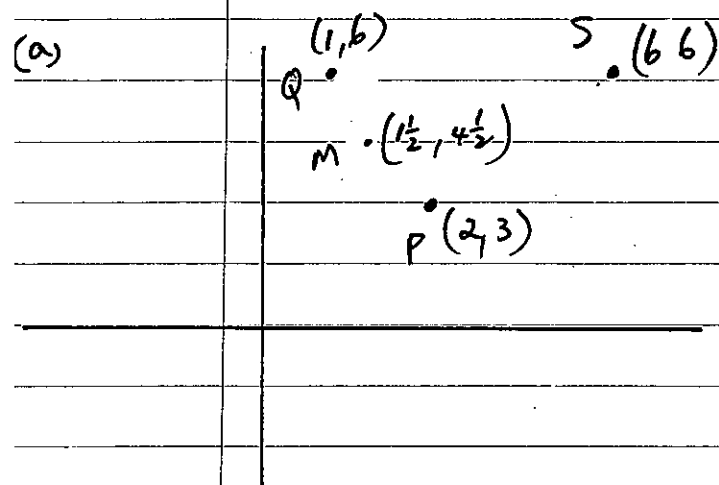
(iii) $y = \log(\cos x)$
 $y' = \frac{1}{\cos x} \times -\sin x$
 $= \frac{-\sin x}{\cos x}$
 $= -\tan x$

(b) $\int_0^{\pi/6} \cos 2x dx = \left[\frac{1}{2} \sin 2x \right]_0^{\pi/6}$
 $= \frac{1}{2} \sin 2 \times \frac{\pi}{6} - \frac{1}{2} \sin 2 \times 0$
 $= \frac{1}{2} \sin \frac{\pi}{3} - \frac{1}{2} \sin 0$
 $= \frac{1}{2} \times \frac{\sqrt{3}}{2} - 0$
 $= \frac{\sqrt{3}}{4} = 0.433$

(c) (i) $\int \sqrt{x} dx = \int x^{\frac{1}{2}} dx$
 $= \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C$
 $= \frac{2}{3} x^{\frac{3}{2}} + C$

(ii) $\int 2x^2 e^{x^3} dx = \frac{2}{3} \int 3x^2 e^{x^3} dx$
 $= \frac{2}{3} e^{x^3} + C$

Question 3.



(i) $M = \left(\frac{1+2}{2}, \frac{3+6}{2} \right)$
 $= \left(1\frac{1}{2}, 4\frac{1}{2} \right)$

(ii) $m_{PQ} = \frac{3-6}{2-1}$
 $= -\frac{3}{1}$
 $\therefore m_1 = -3 \quad m_2 = \frac{1}{3}$

$y - y_1 = m(x - x_1)$

$y - 4\frac{1}{2} = \frac{1}{3} \left(x - 1\frac{1}{2} \right)$

$3y - 13\frac{1}{2} = x - 1\frac{1}{2}$

$x - 3y + 12 = 0$

(iii) $x = 0 \quad \therefore -3y + 12 = 0$
 $3y = 12$
 $y = 4$

(v) $y = 6 \quad x - 3y + 12 = 0$
 $x - 18 + 12 = 0$
 $x = 6$

(1) $PQ = \sqrt{(2-1)^2 + (3-6)^2}$
 $= \sqrt{1^2 + (-3)^2}$
 $= \sqrt{10}$

$MS = \sqrt{\left(6 - 1\frac{1}{2}\right)^2 + \left(6 - 4\frac{1}{2}\right)^2}$
 $= \sqrt{\left(4\frac{1}{2}\right)^2 + \left(1\frac{1}{2}\right)^2}$
 $= \sqrt{22.5}$

$\therefore \text{Area} = \frac{\sqrt{10} \times \sqrt{22.5}}{2}$
 $= \frac{\sqrt{225}}{2}$
 $= \frac{15}{2} \text{ sq. units.}$

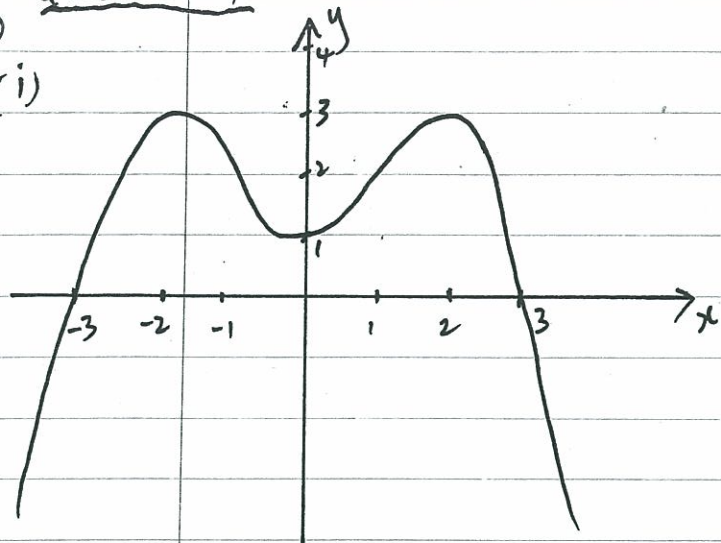
(b) (i) $y = x \sin x$
 $y' = 1 \times \sin x + x \times \cos x$
 $= \sin x + x \cos x$

(ii) $y = \frac{\ln x}{x^2}$
 $y' = \frac{x^2 \times \frac{1}{x} - 2x \ln x}{x^4}$
 $= \frac{x - 2x \ln x}{x^4}$
 $= \frac{x(1 - 2 \ln x)}{x^4 x^3}$
 $= \frac{1 - 2 \ln x}{x^3}$

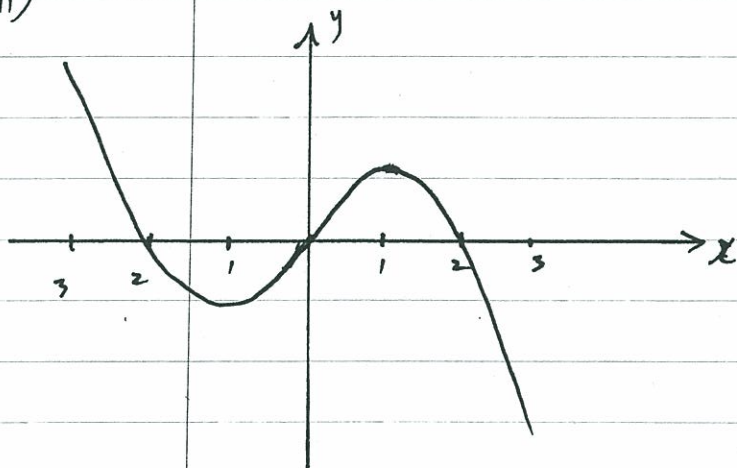
Question 4

a)

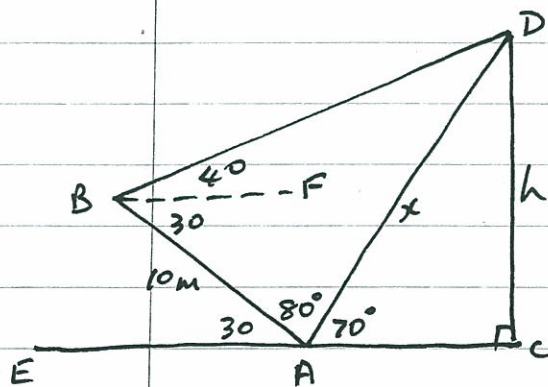
(i)



ii)



b)



BF // EA horizontal lines

$\therefore \angle FBA = \angle EAB$ (alternate angles on parallel lines)

$\therefore \angle FBA = 30^\circ$

$\therefore \angle ABD = 70^\circ$

(ii)

$$\frac{x}{\sin 70} = \frac{10}{\sin 30}$$

$$\therefore x = \frac{10 \sin 70}{\sin 30} = 18.79$$

(iii)

$$\frac{h}{x} = \sin 70$$

$$h = x \sin 70$$

$$= \frac{10 \sin 70}{\sin 30} \times \sin 70$$

$$= 17.66$$

$$= 17.7$$

Question 5

$$\begin{aligned} \text{(a)} \quad \int_0^5 f(x) dx &= \triangle - \square \\ &= \frac{bh}{2} - \frac{\pi r^2}{4} \\ &= \frac{2 \times 3}{2} - \frac{\pi 3^2}{4} \\ &= 3 - \frac{9\pi}{4} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \log_a 24 &= \log_a 3 \times 8 \\ &= \log_a 3 + \log_a 8 \\ &= \log_a 3 + \log_a 2^3 \\ &= \log_a 3 + 3 \log_a 2 \\ &= y + 3x \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \sum_{n=3}^8 2n^2 - 3 &= 15 + 29 + 47 + 69 + 95 + 123 \\ &= 380 \end{aligned}$$

Question 5 Cont.

$$\begin{aligned}
 (b) (i) \int_0^4 x \sqrt{x} dx &= \int_0^4 x \cdot x^{\frac{1}{2}} dx \\
 &= \int_0^4 x^{1\frac{1}{2}} dx \\
 &= \left[\frac{2x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^4 \\
 &= \frac{2}{\frac{3}{2}} \left(4^{\frac{3}{2}} - 0 \right) \\
 &= \frac{2}{\frac{3}{2}} (32 - 0) \\
 &= \frac{64}{3}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \int \frac{5x^2}{x^3+1} dx &= \frac{5}{3} \int \frac{3x^2}{x^3+1} dx \\
 &= \frac{5}{3} \ln(x^3+1) + C
 \end{aligned}$$

$$\begin{aligned}
 (iii) \int_0^2 e^x + e^{-x} dx &= \left[e^x - e^{-x} \right]_0^2 \\
 &= (e^2 - e^{-2}) - (e^0 - e^{-0}) \\
 &= e^2 - \frac{1}{e^2} - (1 - 1) \\
 &= e^2 - \frac{1}{e^2}
 \end{aligned}$$

Question 6.

$$a) e^{2x} + 5e^x - 14 = 0$$

$$\text{let } m = e^x$$

$$\therefore m^2 + 5m - 14 = 0$$

$$(m+7)(m-2) = 0$$

$$m = -7, 2$$

$$\therefore e^x = -7$$

$$e^x = 2$$

no solution

$$x = \log 2$$

$$(b) (i) x^2 = 16y$$

$$4a = 16$$

$$a = 4$$

$\therefore$ directrix is $y = -4$

$$(ii) y = \frac{x^2}{16}$$

$$y' = \frac{2x}{16}$$

$$= \frac{1}{8}x$$

$$\therefore x = 4 \quad m = \frac{4}{8} = \frac{1}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{1}{2}(x - 4)$$

$$2y - 2 = x - 4$$

$$x - 2y - 2 = 0$$

$$(iii) y = -4$$

$$x - 2y - 2 = 0$$

$$x - 2(-4) - 2 = 0$$

$$x + 8 - 2 = 0$$

$$x = -6$$

$$\therefore C = (-6, 4)$$

$$(iv) (a) y = 2 - \frac{x}{4}$$

$$\frac{1}{4}x + y - 2 = 0$$

$$d = \frac{\left| \frac{1}{4}x_1 + y_1 - 2 \right|}{\sqrt{\left(\frac{1}{4}\right)^2 + 1^2}}$$

$$= \frac{\left| \frac{1}{4}(-6) - 4 - 2 \right|}{\sqrt{1\frac{1}{16}}}$$

$$= \frac{7\frac{1}{2}}{\sqrt{1\frac{1}{16}}}$$

$$= \frac{15}{2} \times \frac{4}{\sqrt{17}}$$

$$= \frac{30}{\sqrt{17}}$$

Question 6 cont.

(B)

$$\int_{-8}^4 \left(2 - \frac{x}{4} - \frac{x^2}{16} \right) dx$$

$$= \left[2x - \frac{x^2}{8} - \frac{x^3}{48} \right]_{-8}^4$$

$$= \left(2 \times 4 - \frac{4^2}{8} - \frac{4^3}{48} \right) - \left(-16 - \frac{(-8)^2}{8} - \frac{(-8)^3}{48} \right)$$

$$= 29\frac{1}{3}$$

(ii)

$$\frac{9}{MJ} = \frac{12}{8}$$

$$72 = 12 MJ$$

$$MJ = \frac{72}{12}$$

$$= 6$$

$$\therefore MK = 8 - 6$$

$$= 2$$

(b)

(i)	x	-1	0	1	2	3
	2^x	$\frac{1}{2}$	1	2	4	8

(ii)

$$h = 1$$

$$A \approx \frac{h}{3} \left\{ \frac{1}{2} + 4 \times 1 + 2 \right\} + \frac{h}{3} \left\{ 2 + 4 \times 4 + 8 \right\}$$

$$= \frac{1}{3} \left(6\frac{1}{2} \right) + \frac{1}{3} (26)$$

$$= 10.83$$

(c)

$$(i) \quad x = t^2 - 2t + 7$$

$$t=0 \quad x=7$$

(ii)

$$\dot{x} = 2t - 2$$

$$\dot{x}=0 \quad 2t-2=0$$

$$2t=2$$

$$t=1$$

(iii)

$$t=1 \quad x = 1 - 2 + 7 = 6$$

$$t=4 \quad x = 16 - 8 + 7 = 15$$

in 1st second dist = 1 m

in next 3 seconds 9 m

$\therefore$ total dist. travelled = 10 m.

Question 7

(i) $MN \parallel KL$ given

$$\hat{JMN} = \hat{JKL} \text{ (corresponding angles on parallel lines)}$$

$$\text{similarly } \hat{JNM} = \hat{JLK} \text{ (")}$$

$\hat{J}$ is common

$\therefore \triangle JMN \parallel \triangle JKL$ (equiangular)

Question 8

(a) $3x^2 + 2x + k$

(i) $b^2 - 4ac = 2^2 - 4 \times 3 \times k$
 $= 4 - 12k$

(ii) real roots $b^2 - 4ac > 0$

$$\therefore 4 - 12k > 0$$

$$-12k > -4$$

$$k < \frac{-4}{-12}$$

$$k < \frac{1}{3}$$

b) 3, 2.4, 1.92

(i) $r = 0.8$

$$1.92 \times 0.8 = 1.536 \text{ m}$$

(ii) geometric sequence

(iii) $S_{\infty} = \frac{a}{1-r}$

$$= \frac{2.4}{1-0.8}$$

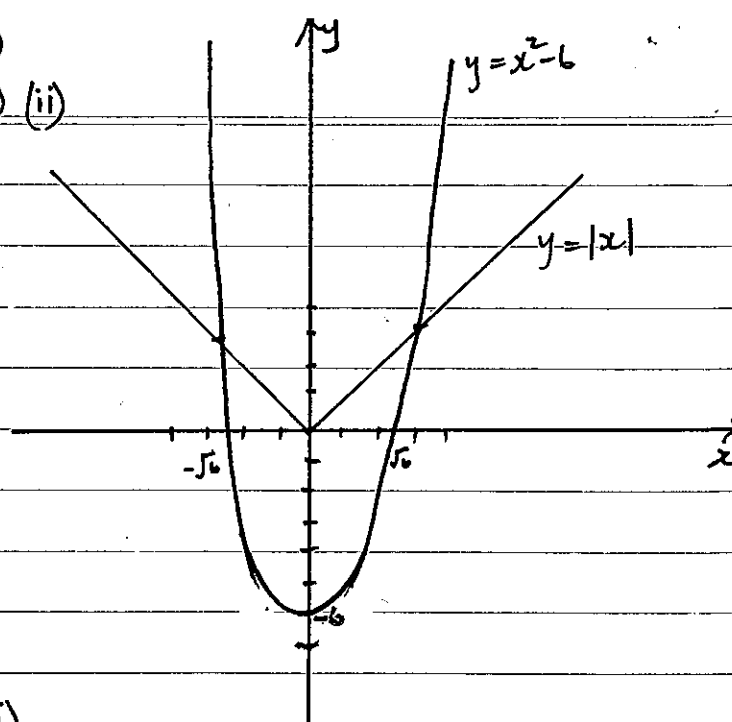
$$= 12$$

$$\text{Total dist} = 3 + 2 \times S_{\infty}$$

$$= 3 + 24$$

$$= 27 \text{ m.}$$

(c)
 (i) (ii)



(iii) $x^2 - 6 = x$

$$x^2 - x - 6 = 0$$

$$(x+2)(x-3) = 0$$

$$x = -2, 3$$

$x = 3$ but by symmetry from graph
 $x = -3$

(iv) $-3 < x < 3$

Question 9

(a)

$$\begin{aligned} \text{(i)} \quad P &= 400 & A_1 &= 400(1+0.005)^{24} \\ r &= \frac{6}{12} & A_2 &= 400(1.005)^{23} \\ &= 0.5\% & A_{24} &= 400(1.005) \\ &= 0.005 \end{aligned}$$

$$a = 400(1.005) \quad r = 1.005 \quad n = 24$$

$$\begin{aligned} S_{24} &= \frac{a(r^n - 1)}{r - 1} \\ &= \frac{400(1.005)(1.005^{24} - 1)}{1.005 - 1} \\ &= \$10223.65 \end{aligned}$$

$$\text{(ii)} \quad 16000 = \frac{P(1.005)(1.005^{24} - 1)}{0.005}$$

$$P = \frac{16000 \times 0.005}{(1.005)(1.005^{24} - 1)}$$

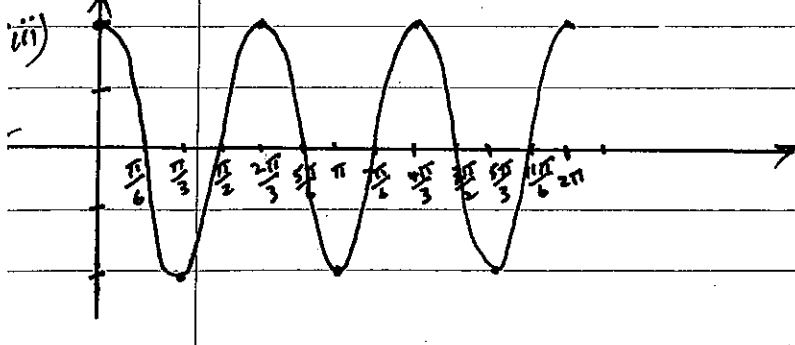
$$= 625.99$$

$\therefore$ Penny needs to save $\approx \$626$

b) $y = 2 \cos 3x$

(i) amp = 2

(ii) period = $\frac{2\pi}{3}$



$$\begin{aligned} \text{(c) (i)} \quad |V_0| &= \pi \int_0^{\frac{\pi}{6}} (\tan 2x)^2 dx \\ &= \pi \int_0^{\frac{\pi}{6}} \tan^2 2x dx. \end{aligned}$$

$$\begin{aligned} \tan^2 \theta + 1 &= \sec^2 \theta \\ \tan^2 \theta &= \sec^2 \theta - 1 \end{aligned}$$

$$\begin{aligned} \therefore \pi \int_0^{\frac{\pi}{6}} \tan^2 2x dx &= \pi \int_0^{\frac{\pi}{6}} (\sec^2 2x - 1) dx \\ \text{(ii)} \quad &= \pi \left(\tan 2x - x \right) \Big|_0^{\frac{\pi}{6}} \\ &= \pi \left[\left(\tan \frac{\pi}{3} - \frac{\pi}{6} \right) - \left(\tan 0 - 0 \right) \right] \\ &= \pi \left(\sqrt{3} - \frac{\pi}{6} \right) \end{aligned}$$

Question 10

$$(a) \quad (x+3)^2 = 4y - 14$$

$$(x+3)^2 = 4(y - 3\frac{1}{2})$$

$$(i) \quad \text{Vertex} = (-3, 3\frac{1}{2})$$

$$(ii) \quad \text{focal length} \quad 4a = 4$$

$$a = 1$$

$$\therefore \text{focus} = (-3, 4\frac{1}{2})$$

$$(iii) \quad \text{directrix} \quad y = 2\frac{1}{2}$$

$$(b) (i) \quad \text{arc length} = r\theta$$

$$\text{semi circle} = \frac{1}{2} \cdot 2\pi r$$

$$\text{but } r = \frac{r}{2} \quad \therefore = \frac{\pi r}{2}$$

$$\therefore P = r + r\theta + \frac{\pi r}{2}$$

$$(ii) \quad \text{Area} = \frac{1}{2} r^2 \theta + \frac{1}{2} \pi \left(\frac{r}{2}\right)^2$$

$$= \frac{1}{2} r^2 \theta + \frac{\pi r^2}{8}$$

$$A = 1$$

$$\therefore \frac{1}{2} r^2 \theta + \frac{\pi r^2}{8} = 1$$

$$\frac{1}{2} r^2 \theta = 1 - \frac{\pi r^2}{8}$$

$$\frac{1}{2} r^2 \theta = \frac{8 - \pi r^2}{8}$$

$$\theta = \frac{8 - \pi r^2}{8} \times \frac{2}{r^2}$$

$$= \frac{8 - \pi r^2}{4r^2}$$

$$\therefore \theta = \frac{8}{4r^2} - \frac{\pi r^2}{4r^2}$$

$$\theta = \frac{2}{r^2} - \frac{\pi}{4}$$

$$\text{but } P = r + r\theta + \frac{\pi r}{2}$$

$$\therefore P = r + r\left(\frac{2}{r^2} - \frac{\pi}{4}\right) + \frac{\pi r}{2}$$

$$= r + \frac{2}{r} - \frac{\pi}{4}r + \frac{\pi}{2}r$$

$$= r + \frac{2}{r} + \frac{\pi}{4}r$$

$$P = \frac{2}{r} + r\left(1 + \frac{\pi}{4}\right)$$

$$(iii) \quad P = 2r^{-1} + \left(1 + \frac{\pi}{4}\right)r$$

$$P' = -2r^{-2} + 1 + \frac{\pi}{4}$$

$$= \frac{-2}{r^2} + 1 + \frac{\pi}{4}$$

$$P' = 0 \quad \text{for min}$$

$$P'' = -4r^{-3}$$

$$= \frac{-4}{r^3}$$

$$\therefore \frac{-2}{r^2} + 1 + \frac{\pi}{4} = 0$$

$\therefore$ for min

$$\frac{2}{r^2} = 1 + \frac{\pi}{4}$$

$$2 = \left(1 + \frac{\pi}{4}\right)r^2 \quad r = \pm \sqrt{\frac{8}{4+\pi}}$$

$$r^2 = \frac{2}{1 + \frac{\pi}{4}}$$

$$r^2 = \frac{2}{\frac{4+\pi}{4}}$$

$$= \frac{8}{4+\pi}$$

$$r = \pm \sqrt{\frac{8}{4+\pi}}$$

Question 10 cont.

$$\theta = \frac{2}{r^2} - \frac{\pi}{4}$$

$$= \frac{2}{\frac{8}{4+\pi}} - \frac{\pi}{4}$$

$$= \frac{2(4+\pi)}{8} - \frac{\pi}{4}$$

$$= \frac{4}{4} + \frac{\pi}{4} - \frac{\pi}{4}$$

$$= 1 \text{ radian}$$